

Title: Task 5: Ensemble Learning, Optimization & Real-World ML Pipeline

Objective: To build a complete classification system, compare ensemble learning techniques against a baseline model, and optimize performance using hyperparameter tuning.

Model Comparison

- **Logistic Regression (Baseline):** The baseline linear model achieved high initial accuracy but is inherently limited by its assumption of linear relationships between features. It relies heavily on strict data scaling for optimal performance.
- **Random Forest:** This ensemble model successfully utilized bagging (bootstrap aggregating) to combine multiple decision trees. It handled the high-dimensional feature space of the dataset exceptionally well, reducing the overfitting typically seen in single decision trees.
- **Gradient Boosting:** By building trees sequentially to correct the errors of preceding weak learners, Gradient Boosting demonstrated aggressive optimization. It yielded highly competitive precision and recall, though it required longer training time compared to the Random Forest.

Optimization Explanation

- To improve model performance for production, GridSearchCV was implemented to automate hyperparameter tuning.
- The grid evaluated constraints on the Random Forest, specifically testing the number of trees (n_estimators) and the maximum depth of the trees (max_depth).
- Coupled with 3-fold cross-validation, this process systematically identified the optimal parameter combination to maximize testing accuracy while preventing training data memorization.

Final Decision The **Tuned Random Forest Classifier** is selected as the final production model. While Gradient Boosting is powerful, the Tuned Random Forest provides a superior balance of performance, stability, and interpretability. It requires less extensive hyperparameter tuning to prevent overfitting, handles the unscaled feature data robustly, and the grid search optimization guarantees it operates at peak efficiency for this specific classification task.